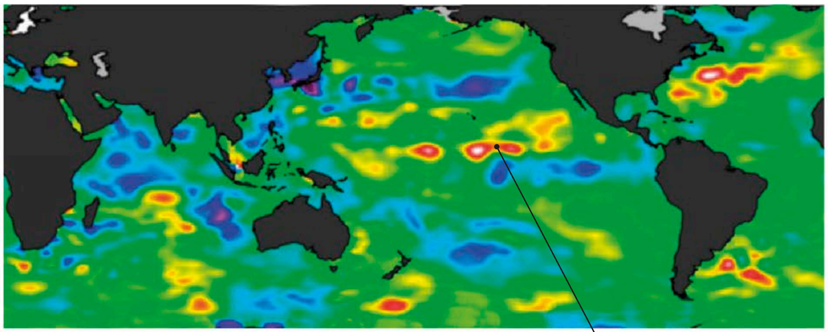
WEATHER WATCHERS

The first weather satellite, TIROS-1, was launched in 1960, and by the 1970s, satellites were placed in high orbits where they could photograph weather patterns across large areas of Earth. Today, remote-sensing instruments can monitor wind speeds, temperature, and other conditions in the atmosphere and oceans, measuring large-scale patterns and changes to Earth's overall climate.

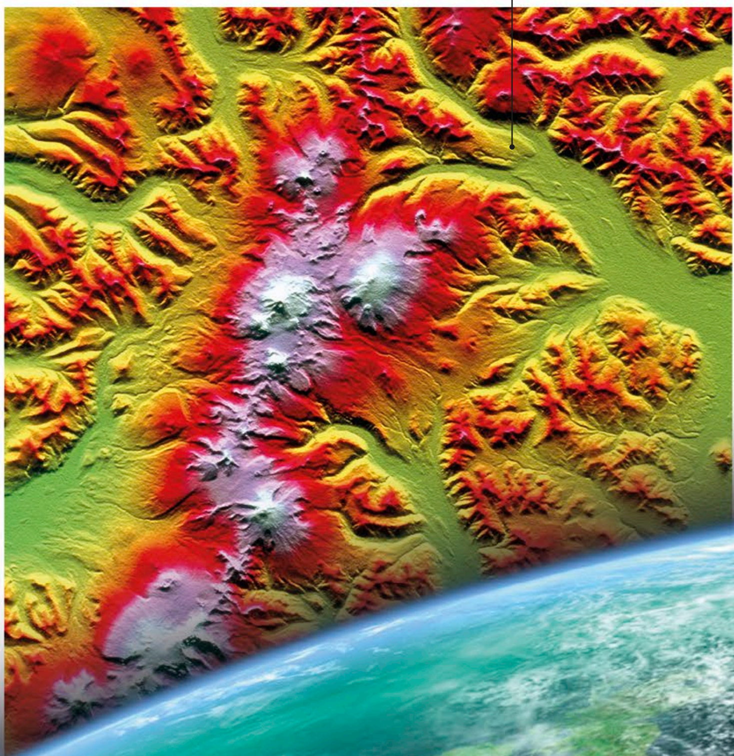


Satellite map of ocean temperatures during the 2017 El Niño climate event

False color shows central Pacific Ocean waters are warmer than usual.

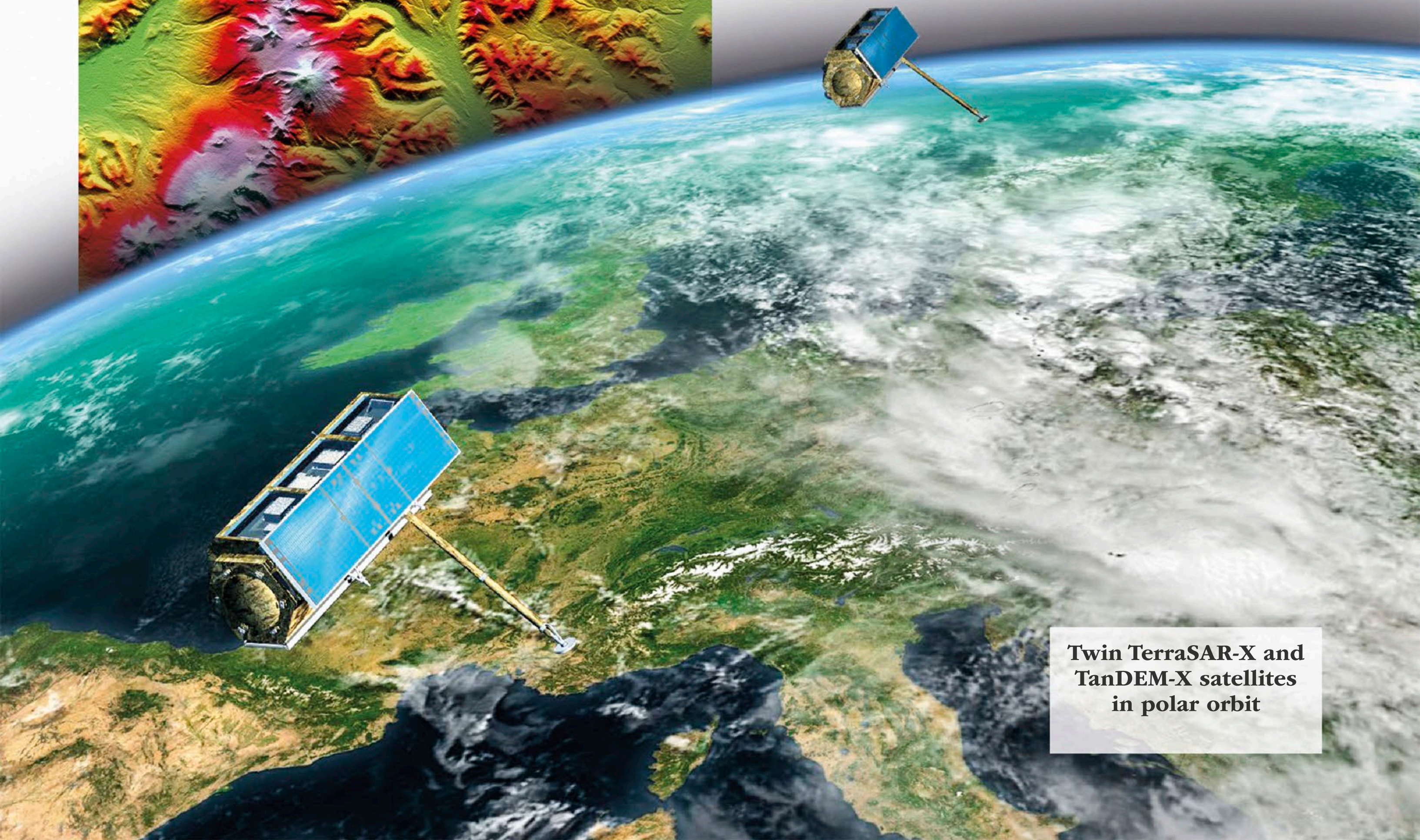
▼ RADAR COMPOSITE
Satellite radar can help map details of features in inaccessible areas, such as these Russian volcanoes, in any weather, day or night.

Color-coding reveals elevation of the peaks: green is lowest, rising through yellow, red, and pink to white, the highest.



RADAR MAPPING

Modern radar satellites map Earth's surface with high precision by firing beams of radio waves at the surface and measuring their reflections. The time the signal takes to return indicates the distance to the reflecting surface, while changes to the reflected waves reveal other details such as surface texture and mineral composition. The German space agency's TanDEM-X mission used two satellites to create the most detailed three-dimensional map of Earth so far.



Twin TerraSAR-X and TanDEM-X satellites in polar orbit